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Stock specific relative abundance of Columbia River juvenile Chinook salmon off the Southeast Alaska coast

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ABSTRACT

Migration patterns of juvenile Columbia River Chinook salmon (*Oncorhynchus tshawytscha*) differ among stocks and life history types, creating diverse marine distributions of these fish. This results in different stocks being subject to different ocean conditions during their first summer of marine residence, a time that is critical for their survival. Understanding their early marine distributions, and the conditions that affect their survival, may enhance conservation efforts for these stocks, many of which are protected under the Endangered Species Act. We analyzed juvenile Chinook salmon samples collected in trawls made from 2011 to 2015 off the southeastern (SE) Alaskan panhandle, and off the coasts of Washington and Oregon. We used genetic stock identification techniques to estimate stock proportions of juvenile Chinook salmon in each of these areas. Results indicated that most juvenile Chinook salmon in our SE Alaska coast study area in July originate from Columbia River spring-run stocks. We found a significant relationship in catch per unit effort (CPUE) over all stocks between the SE Alaska and the Washington and Oregon coast samples. In 2011, CPUE for Columbia River spring-run stocks for the SE Alaska coast was lower than for the Washington and Oregon coast, suggesting a differing level of marine mortality that year. We also found a significant relationship between juvenile CPUE of interior Columbia River spring-run stocks off the SE Alaska coast and adult counts at Bonneville Dam two years later. Our results provide marine life history, performance and survival information that supports management and recovery efforts for Columbia River Chinook salmon.

1. Introduction

Several studies have shown that some Columbia River Chinook salmon (*Oncorhynchus tshawytscha*) migrate rapidly north after entering the ocean, and that some smolts reach Alaskan marine waters within a few weeks (Orsi et al., 2000; Trudel et al., 2009; Fisher et al., 2014; Rechisky et al., 2014). However, this behavior is not universal, and migration patterns differ greatly among Columbia River stocks and among juvenile life history types (Hartt and Dell, 1986; Fisher and Pearcy, 1995; Trudel et al., 2009; Tucker et al., 2011). As a result, during their first summer of marine residence, Columbia River Chinook salmon juveniles are distributed in coastal areas from Oregon to Alaska (Tucker et al., 2011; Fisher et al., 2014; Teel et al., 2015). These diverse marine distributions are potentially important because varying ocean conditions in differing marine provinces might differentially affect salmon survival (Mueter et al., 2005; Pyper et al., 2005). However, descriptions of Columbia River Chinook salmon ocean migration patterns are incomplete, as is an understanding of how oceanic regimes regulate marine survival. Filling these knowledge gaps will enhance conservation efforts for Columbia River Chinook salmon, which currently have five evolutionarily significant units (ESUs) protected under the U.S. Endangered Species Act (ESA) (Ford et al., 2015).

Many studies of salmon marine distribution have relied on physical tags placed in hatchery fish prior to ocean entry (Orsi and Jaenicke, 1996; Trudel et al., 2009; Fisher et al., 2014). Fish identifications using tags are accurate, and together with information on hatchery release date can be used to estimate migration rates. However, this approach can only work with fish that are tagged and their tags recaptured. Thus,

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the majority of juvenile salmon collected in an ocean survey provide no information on where the fish are from. Stock identification using genetic data obtained from both ocean-caught fish and samples from potential source populations (a "genetic baseline") has also been widely used to study the marine distributions of juvenile salmon (Teel et al., 2003; Van Doornik et al., 2007; Tucker et al., 2009; Seeb et al., 2011; Beacham et al., 2012, 2014; Hassrick et al., 2016). Because all fish sampled in a survey provide genetic information, genetic studies are particularly useful for estimating overall stock composition, relative abundance of each stock (i.e. stock-specific catch per unit effort, CPUE), and the stock origin of individual fish. The accuracy and resolution of these estimates are dependent on the level of genetic differentiation among the source populations and also require the existence of a geographically comprehensive genetic baseline dataset.

Studies of salmon in the ocean are frequently fragmented by geographic and political boundaries because fisheries agencies and research institutions often prioritize study of local fish in local waters (Brodeur et al., 2000). These limitations have hampered broad-scale comprehension of Columbia River Chinook salmon that rapidly transit from U.S. waters to Canadian waters and then back to U.S. waters during late spring and summer migrations that encompass 1000s of kilometers of coastal ocean habitat. Similarly, genetics laboratories in different jurisdictions often have divergent research priorities, thereby impeding the standardization of DNA baseline data that are needed to delineate stock compositions of ocean-caught salmon (Moran et al., 2006). Herein we take advantage of two recent developments: 1) concurrent marine juvenile salmon surveys conducted in June off the Washington and Oregon coasts, and in July off the SE Alaska coast, 2011–2015; and 2) genetic analysis of Chinook salmon from both surveys using an internationally standardized, coast-wide baseline that supports identification of fish from the Columbia River. This information provides the opportunity for greater understanding of potential differences in marine survival rates among stocks, including fish from imperiled ESUs that are currently protected under the ESA.

2. Materials and methods

2.1. Field survey sampling

Juvenile Chinook salmon were captured over the continental shelf, in the Gulf of Alaska, and off the SE Alaska coast in July 2011–2015, and September 2013, as part of the Gulf of Alaska Integrated Ecosystem Research Program (Dickson and Baker, 2016). Samples were collected using a pelagic rope trawl (Cantrawl model #400) modified to fish at the surface, at gridded survey stations arranged along a series of transects that extended from nearshore to offshore (Fig. 1). The Cantrawl is a 198-m long mid-water rope trawl with hexagonal mesh wings and body and a 1.2 cm mesh liner in the codend. It was towed at an average of 2.7 km h^{-1}, had an average horizontal spread of 42.2 m, and an average vertical spread of 37.1 m. All tows lasted 30 min and covered an average distance of 1.4 km. At each station, CPUE (#/km^2) was calculated by dividing the number of juvenile Chinook salmon captured in a tow by the product of the haversine distance (km) and the horizontal spread (km).

After the trawl was retrieved, the catch was emptied onto a sorting table where fish were sorted according to species and age-class. A fin clip was taken from the interior edge of the pectoral fin and preserved in 100% ethanol. Fish were frozen at sea, and later were thawed in a laboratory where fork length (mm) and weight (g) were recorded. All juvenile Chinook salmon taken in our trawl surveys and used in our analyses were assumed to be yearlings (fish that had spent one year in freshwater before migrating to sea) based upon their fork lengths. Fish were identified as having a hatchery origin if their adipose fin had been removed, or if they had been tagged with a coded wire tag (CWT).

To compare stock proportions of juvenile Chinook salmon off the SE Alaska coast and their location when they first enter the ocean, we took advantage of data that have been collected from a similar sampling effort off the Washington and Oregon coasts (Brodeur et al., 2000). That sampling survey has been conducted every year in June, since 1998. Juvenile Chinook salmon samples were collected from these trawls as described by Teel et al. (2015). The CPUE was calculated for each year by dividing the number of juvenile Chinook salmon captured by the distance the net was trawled (#/km). There was no measure of net mensuaration. While the CPUEs for these two surveys are not directly comparable due to differing methodologies, they could be compared for relative variation among years within locations.

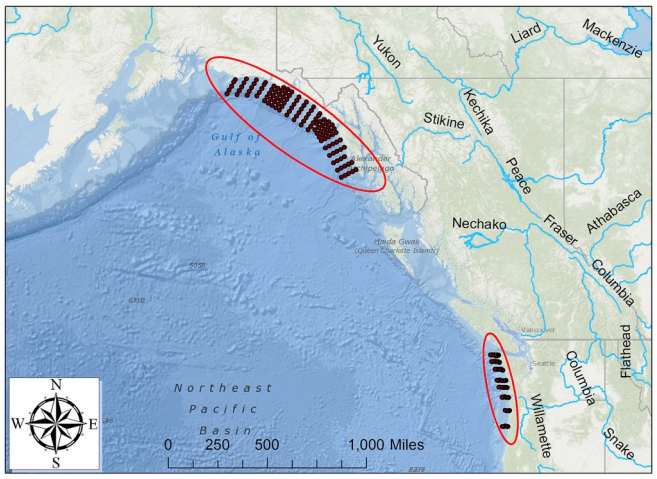


Fig. 1. Locations of survey trawls off the SE Alaska, Washington, and Oregon coasts from which juvenile Chinook salmon (*Oncorhynchus tshawytscha*) samples were obtained.

2.2. Genetic data

Genomic DNA was extracted from fin tissue samples by using Wizard Genomic DNA Purification Kits (Promega Corp.) according to the manufacturer's protocols. Polymerase chain reactions (PCRs) were then used to amplify 13 microsatellite DNA loci that have been standardized for Chinook salmon by several West Coast salmon genetics laboratories (Seeb et al., 2007). We then used the Applied Biosystems Inc. (ABI) 3100 or 3700 genotyping platform to size fractionate the PCR products. Genotypes were determined through the use of ABI GeneScan, Genotyper, and Genemapper software.

Estimates of the stock of origin for each fish sampled were made using genetic stock identification (GSI). This method makes use of a baseline of genetic information collected from samples of known origin. The baseline we used consisted of genotypes collected from over 22,000 individual fish, representing 163 populations from 15 regional groups (genetic stocks) ranging from Alaska to Central California (Seeb et al., 2007; Moran et al., 2013; Hess et al., 2014; available at: www.fishgen.net). Stock origin estimates for individual fish were calculated using the program ONCOR (Kalinowski, 2007), which makes use of the likelihood model of Rannala and Mountain (1997).

The Columbia River genetic stocks identified in our analysis (Table 1) were previously described by Johnson et al. (2013) and Teel et al. (2015). Briefly, yearling smolts originate from several genetically distinct stocks, each described by its historical spawning distribution and the season adults return to freshwater, known as "run-time" (Waples et al., 2004). Nearly 32 million yearlings are released from hatcheries each year, with the majority from facilities that propagate Snake River and Middle/Upper Columbia River interior basin spring-run stocks (Regional Mark Information System, www.rmpc.org). Releases of hatchery yearlings from lower river spring-run stocks are also substantial, primarily from the Willamette River stock in Oregon and the Western Cascade stock, including the Lewis and Cowlitz Rivers in

Table 1
Sample sizes and genetic stock percentages of juvenile Chinook salmon caught in SE Alaska coast sampling trawls, and the overall percentage of fish from each stock that had been marked with an adipose fin clip and/or CWT. Genetic stocks highlighted in bold are those used in subsequent comparative analyses.

	July							September	
	2011	2012	2013	2014	2015	All years	Marked	2013	Marked
Total Sample Size	128	212	203	167	118	828	700	47	32
Genetic Stock Percentages									
SE Alaska/Transboundary Area	14.8	6.1	12.8	13.2	6.8	10.6	17.4	12.8	0.0
Skeena R./Nass R.	1.6	0.0	0.0	0.6	0.0	0.4	100.0	2.1	100.0
BC Coast/Vancouver I./Strait of Georgia	7.0	0.9	0.0	1.2	5.1	2.3	30.0	0.0	n/a
Fraser R.	3.9	0.9	1.5	1.2	5.1	2.2	30.8	0.0	n/a
Puget Sound/Strait of Juan de Fuca	4.7	0.5	1.5	0.0	3.4	1.7	25.0	0.0	n/a
Washington Coast	0.8	0.0	1.5	0.0	1.7	0.7	60.0	0.0	n/a
West Cascade spring	3.9	2.8	4.4	5.4	3.4	4.0	82.1	0.0	n/a
Lower/Middle Columbia R. fall-run	6.3	1.4	2.0	0.6	2.5	2.3	54.5	0.0	n/a
Willamette R. spring-run	**10.2**	**8.0**	**14.3**	**4.2**	**13.6**	**9.9**	**85.5**	**0.0**	**n/a**
Middle/Upper Columbia R. spring-run	**6.3**	**35.8**	**21.7**	**32.9**	**26.3**	**25.8**	**85.0**	**53.2**	**72.0**
Upper Columbia R. summer-/fall-run	**22.7**	**1.9**	**3.9**	**4.8**	**7.6**	**7.0**	**72.4**	**4.3**	**100.0**
Snake R. spring-run	**7.0**	**38.7**	**32.5**	**34.7**	**22.0**	**29.1**	**79.3**	**27.7**	**84.6**
Snake R. fall-run	3.1	0.9	1.5	0.0	1.7	1.3	71.4	0.0	n/a
North and Central Oregon Coast	7.8	1.9	2.5	1.2	0.8	2.7	41.7	0.0	n/a
South Oregon Coast/California	0.0	0.0	0.0	0.0	0.0	0.0	n/a	0.0	n/a

Washington. Yearling smolts from fall- and summer-run populations are also produced by hatcheries, nearly all of which use Upper Columbia River or Snake River interior basin stocks. In addition, some yearling smolts are produced from other fall-run stocks originating farther downstream than the fall-run stocks described above. We report a single combined estimate of fall-run fish's contributions to our catches (Lower/Middle Columbia River fall-run) as relatively few yearling smolts are released from these stocks.

We only conducted detailed analyses on Columbia River yearling stocks that averaged at least 5% of the catch in the SE Alaska coast surveys.

2.3. Data analyses

We compared average CPUE values of Columbia River stocks found off the SE Alaska coast to those found off the Washington and Oregon coasts using a simple linear regression. This allowed us to make inferences about differences in migration timing and pathways among these stocks. We also compared the CPUE off the SE Alaska coast of the middle and upper Columbia River spring-run stock and Snake River spring-run stock to the number of adult returns for those stocks two years later, as counted at Bonneville Dam on the Columbia River (Fish Passage Center, www.fpc.org). Assuming an average return age of four years (Healey, 1991), this allowed us to test for a relationship between those stocks' relative abundance off the SE Alaska coast during their first summer at sea as compared to their abundance when returning as adults to natal freshwater streams.

3. Results

3.1. SE Alaska coast stock composition

The numbers of juvenile Chinook salmon sampled off the SE Alaska coast in July 2011–2015 were 128, 212, 203, 167, and 118, respectively (Table 1). The single collection from September 2013 numbered 47. Juvenile Chinook salmon captured in the July SE Alaska coast trawls were found to have originated from 14 of the 15 stock groups, but they were predominantly from the Columbia River (Table 1). Overall, Columbia River stocks were estimated to comprise 79.5% of the samples. The largest contributors were the Snake River spring-run (29.1%) and Middle/Upper Columbia River spring-run (25.8%) stocks, both of which originate in the interior Columbia River Basin. Other Columbia River stocks included Willamette River spring-run (9.9%) and Upper Columbia River summer-/fall-run (7.0%). The largest contribution from

Table 2
Sample sizes and genetic stock percentages of juvenile Chinook salmon caught in Washington and Oregon coastal sampling trawls in June, and the overall percentage of fish from each stock that had been marked with an adipose fin clip and/or CWT. Genetic stocks highlighted in bold are those used in subsequent comparative analyses.

	2011	2012	2013	2014	2015	All years	Marked
Total Sample Size	311	199	214	140	154	1018	868
Genetic Stock Percentages							
SE Alaska/Transboundary Area	0.0	0.0	0.0	0.0	0.0	0.0	n/a
Skeena R./Nass R.	0.0	0.0	0.0	0.0	0.0	0.0	n/a
BC Coast/Vancouver I./Strait of Georgia	0.0	0.0	0.0	0.0	0.0	0.0	n/a
Fraser R.	0.0	0.0	0.0	0.0	0.0	0.0	n/a
Puget Sound/Strait of Juan de Fuca	2.5	0.9	0.0	0.0	0.0	0.6	100.0
Washington Coast	1.0	0.5	0.0	0.0	0.0	0.3	100.0
West Cascade spring	5.0	5.2	10.7	37.1	21.5	14.9	94.5
Lower/Middle Columbia R. fall-run	2.1	4.4	7.9	1.4	1.1	4.1	69.6
Willamette R. spring-run	**9.1**	**5.5**	**6.5**	**7.9**	**15.1**	**7.9**	**79.2**
Middle/Upper Columbia R. spring-run	**12.6**	**19.2**	**15.4**	**2.1**	**8.6**	**12.7**	**84.0**
Upper Columbia R. summer-/fall-run	**39.2**	**34.1**	**28.5**	**24.3**	**23.7**	**30.0**	**90.0**
Snake R. spring-run	**19.3**	**22.6**	**20.6**	**11.4**	**21.5**	**19.3**	**86.9**
Snake R. fall-run	9.2	7.6	10.3	15.7	6.5	10.0	78.0
North and Central Oregon Coast	0.0	0.0	0.0	0.0	1.1	0.1	33.3
South Oregon Coast/California	0.0	0.0	0.0	0.0	1.1	0.1	0.0

Table 3
Catch per unit effort (CPUE) of Chinook salmon originating from Columbia River stocks caught in our sampling trawls in two regions. Note that due to differing methodologies, the values from SE Alaska are number of fish per square kilometer, whereas the Washington (WA) and Oregon (OR) coast values are number of fish per kilometer.

Sampling region	Genetic stock	2011	2012	2013	2014	2015
SE Alaska (#/km^2)	West Cascade spring	1.056	1.482	3.306	2.455	1.460
	Lower/Middle Columbia R. fall-run	1.689	0.740	1.469	0.270	1.095
	Willamette R. spring-run	2.745	4.198	10.653	1.909	5.839
	Middle/Upper Columbia R. spring-run	1.689	18.769	16.163	15.002	11.312
	Upper Columbia R. summer-/fall-run	6.123	0.988	2.939	2.182	3.284
	Snake R. spring-run	1.900	20.25	24.245	15.82	9.488
	Snake R. fall-run	0.845	0.494	1.102	0.000	0.730
	Average	2.292	6.703	8.554	5.377	4.744
WA & OR Coast (#/km)	West Cascade spring	0.020	0.068	0.148	0.319	0.120
	Lower/Middle Columbia R. fall-run	0.010	0.058	0.109	0.012	0.060
	Willamette R. spring-run	0.042	0.073	0.090	0.067	0.084
	Middle/Upper Columbia R. spring-run	0.059	0.253	0.213	0.018	0.048
	Upper Columbia R. summer-/fall-run	0.182	0.451	0.393	0.208	0.132
	Snake R. spring-run	0.090	0.299	0.284	0.098	0.120
	Snake R. fall-run	0.043	0.100	0.142	0.135	0.036
	Average	0.449	1.302	1.379	0.857	0.600

non-Columbia River sources was from the Southeast Alaska/Transboundary group (10.6%). Stock composition varied among years, with the greatest shift occurring in 2011, when lower contributions were estimated for Snake River spring-run (7.0%) and Middle/Upper Columbia River spring-run (6.3%) stocks, and a much greater percentage of Upper Columbia River summer-/fall-run fish (22.7%). The single sample from September, collected in 2013, was also predominately from the Columbia River (85.2%), and again, primarily from the Snake River spring-run (27.7%) and Middle/Upper Columbia River spring-run (53.2%) stocks (Table 1). The majority of the fish caught from each of the Columbia River stocks had been marked, and thus, had originated from a hatchery.

The CPUE varied by both stock and year (Table 3; Fig. 2). Snake River spring-run and Middle/Upper Columbia River spring-run stock catch was lowest in 2011, highest in 2012 and 2013 and intermediate in 2014 and 2015. In general, these fish were 5–10 fold more abundant than Upper Columbia River summer-/fall-run or Willamette River spring-run. Our sample catches of Upper Columbia River summer-/fall-run Chinook salmon were relatively low in all years, however, these were the most abundant fish in 2011. Catch of Willamette River spring-run was similar to those of Upper Columbia River summer-/fall-run in most years, with the exception of 2013 when the relative abundance of Willamette River spring-run fish was 2–5 fold greater than observed in other years.

3.2. SE Alaska coast versus Washington and Oregon coast stock composition

Similar to our results from the SE Alaska trawls, the Washington and Oregon coast trawls mostly caught Chinook salmon from Columbia River stocks (Table 2). We found a significant relationship between the average CPUE of Chinook salmon caught off the Washington and Oregon coasts (Table 3) and the CPUE of those caught off SE Alaska ($F = 23.0$, $P = 0.02$, $r^2 = 0.89$, N = 5, Fig. 3). Overall CPUE was lowest in 2011 and highest in 2013 in each region. As total release of yearling hatchery Chinook salmon varied little across these years, the interannual variation in CPUE suggests interannual variation in marine survival.

Correlations within stocks, across years and regions were not significant (Fig. 4). However, a number of patterns were apparent. Snake River spring-run and Middle/Upper Columbia River spring-run Chinook salmon catches varied similarly and were positively correlated across regions for 2012–2015. For 2011, CPUE was relatively much lower off the SE Alaska coast than for the Washington and Oregon coasts for these stocks, suggesting a differing pattern of marine mortality in 2011.

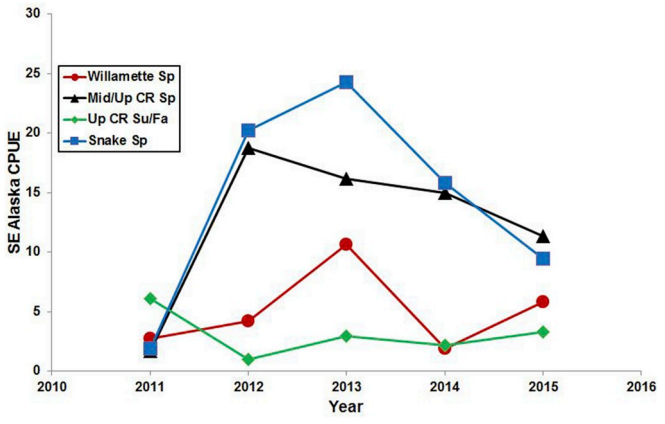


Fig. 2. Catch per unit effort (CPUE) of yearling Columbia River Chinook salmon stocks sampled in SE Alaska over a five year period. Stock groups include Willamette River spring-run (Willamette Sp), Middle/Upper Columbia River spring-run (Mid/Up UCR Sp), Upper Columbia River summer-/fall-run (Up CR Su/Fa), and Snake River spring-run (Snake Sp).

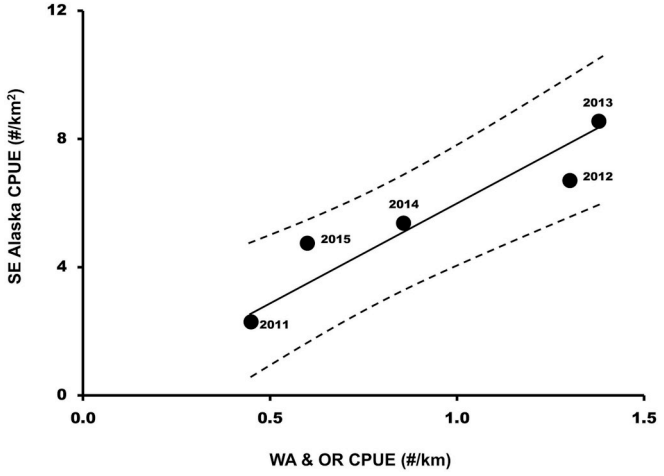


Fig. 3. Average annual catch per unit effort (CPUE) of Columbia River Chinook salmon stocks sampled in SE Alaska versus those sampled off the Washington and Oregon coast over a five year period. Dashed lines are 95% confidence intervals.

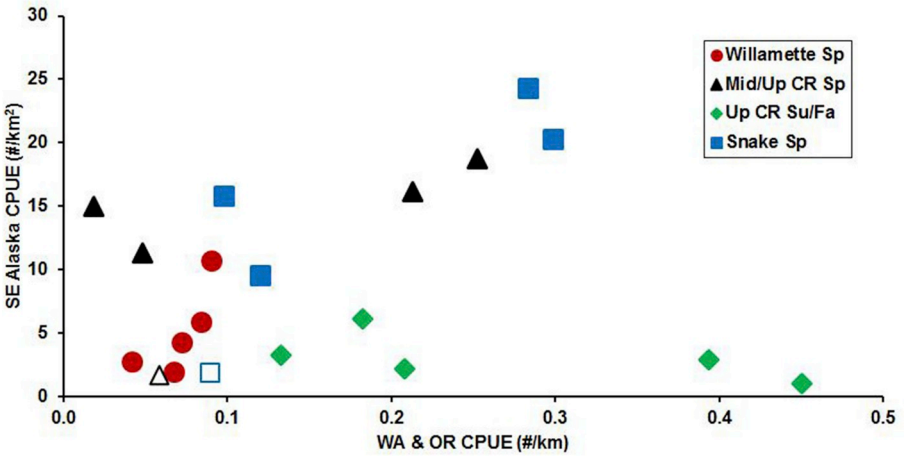


Fig. 4. Catch per unit effort (CPUE) of Columbia River Chinook salmon stocks sampled in the SE Alaska versus those sampled off the Washington and Oregon coast over a 5-year period. Stock groups include Willamette River spring-run (Willamette Sp), Middle/Upper Columbia River spring-run (Mid/Up UCR Sp), Upper Columbia River summer-/fall-run (Up CR Su/Fa), and Snake River spring-run (Snake Sp). Data points from 2011 for Mid/Up CR Sp and Snake Sp are shown as unfilled shapes.

There was relatively little variance in the catch of Willamette River spring-run Chinook salmon off the Washington and Oregon coast (2-fold) as compared to the SE Alaska coast (5-fold), providing little power to determine patterns of relative abundance across regions. A similar pattern was found for Upper Columbia River summer-/fall-run fish, with less variation in catch off the Washington and Oregon coast (3-fold) and greater variation in catch across years off the SE Alaska coast (5-fold).

3.3. Relationship between SE Alaska coast CPUE and spawner abundance

Simple counts of adult spring-run Chinook salmon are made at Bonneville Dam, the dam closest to the Columbia River mouth. These counts combine both Snake River and Middle/Upper Columbia River spring-run Chinook adults on their spawning migration. These counts do not discriminate fish by age, but the majority of fish in any given year are age four, returning to freshwater after two winters in the ocean (1.2). We found a strong positive relationship ($r^2 = 0.79$, $P = 0.04$) between the CPUE of juvenile interior Columbia River spring-run stocks off the SE Alaska coast and adult counts at Bonneville Dam two years later (Fig. 5). This relationship suggests that marine survival of Snake River spring-run and Middle/Upper Columbia River spring-run Chinook adults was mostly determined by the time the juvenile fish reached the SE Alaska coast in 2011–2015.

4. Discussion

The most salient finding of our study was that in July, juvenile Chinook salmon off the SE Alaska coast are predominantly from the Columbia River. Columbia River juveniles comprised nearly 80% of our samples, which were collected during 5 years of sampling in July along the Southeast Alaskan coast. The stock composition estimates highlight the Columbia River basin as a major source of juvenile Chinook salmon in the Northeast Pacific and complement the well-documented contributions of the basin's adult Chinook salmon to commercial and recreational fisheries in Alaska and British Columbia (Beacham et al., 2006; Gilk-Baumer et al., 2013; Weitkamp et al., 2015).

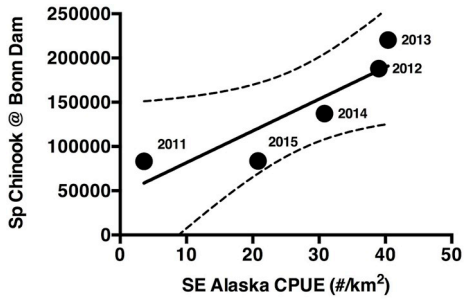


Fig. 5. Catch per unit effort (CPUE) of Columbia River spring-run Chinook salmon in SE Alaska from 2011 to 2015 versus returning Columbia River spring-run Chinook salmon adult counts made two years later at Bonneville Dam on the Columbia River. Dashed lines are 95% confidence intervals.

Our SE Alaska coast results also confirm three major early marine residence patterns, previously described for Columbia River Chinook salmon. First, based on hatchery markings, the majority of the fish in our samples were from the river's hatcheries, a finding consistent with the predominance of hatchery origin fish in yearling populations in marine areas off of Washington and Oregon (Daly et al., 2012; Teel et al., 2015). Second, the absence of subyearling juveniles in our catches is in agreement with previous studies showing that those individuals typically rear for several months in coastal regions near the Columbia River prior to dispersing more broadly (Miller et al., 1983; Fisher and Pearcy, 1995; Trudel et al., 2009; Tucker et al., 2011; Fisher et al., 2014; Teel et al., 2015). And third, our results support the observations of the rapid migration of some Columbia River Chinook salmon yearlings to Alaskan waters, a pattern initially suggested by Hartt and Dell (1986) and subsequently documented in several other studies (Orsi and Jaenicke, 1996; Orsi et al., 2000; Trudel et al., 2009; Tucker et al., 2011; Rechisky et al., 2013; Fisher et al., 2014). Our results, from a study conducted approximately 2200 ocean kilometers from the Columbia River mouth and only about two months following peak yearling ocean entry, emphasizes the length, speed and prevalence of this migration.

However, not all Columbia River yearlings migrate quickly north and our findings reflect the life history variability inherent in Chinook salmon, both within and between stocks. The majority of yearlings caught in our SE Alaska coast surveys were from interior Columbia River spring-run populations, which have the fastest and nearly unimodal northward marine dispersal pattern (Trudel et al., 2009; Fisher et al., 2014). By summer, interior spring run yearlings are rarely found in coastal areas south of Alaska (Fisher et al., 2014). In contrast, our smaller catches of yearlings from Columbia River fall-run populations, and spring-run fish from the lower Columbia River (e.g. Willamette River), are consistent with the diverse dispersal patterns those stocks exhibit. While the majority of those fish disperse north following ocean entry, some individuals also move southward, and by summer are broadly distributed from Oregon to Alaska (Trudel et al., 2009; Fisher et al., 2014; Teel et al., 2015).

We also observed stock-specific and interannual differences in the marine relative abundance patterns of Columbia River stocks. For example, CPUE for Snake River and Mid/Upper Columbia River spring-run Chinook salmon varied 10-fold across years. Yet, the numbers of hatchery fish released from these stocks were consistent during these

same years (Fish Passage Center, www.fpc.org), suggesting that inter-annual variability in riverine or marine survival were responsible for the fluctuations in relative abundance. In contrast, the CPUE of Upper Columbia River summer/fall-run yearlings was relatively stable during our study, although at a lower level than the three most abundant spring-run stocks. As both spring-run and summer/fall-run yearling Chinook salmon enter the ocean at the same location and at about the same time, these data suggest that variable factors related to ocean ecology may have strong stock-specific effects on interannual relative abundance patterns. Indeed, several Columbia River Chinook salmon marine survival studies have shown that interannual smolt to adult survival patterns differ among these stocks (Sharma et al., 2013; Kilduff et al., 2014).

Stock-specific relative abundances of Columbia River Chinook salmon smolts in June off the Washington and Oregon coast and in July off the SE Alaska coast were generally related, with a notable exception (Fig. 4). In 2011 the correlation between the relative abundance of spring-run Chinook salmon off of Washington and Oregon and off the SE Alaska coast was disrupted, with substantially fewer smolts found off the SE Alaska coast than would be predicted based on the 2012–2015 pattern. These data suggest that, compared to other years, greater smolt mortality occurred in 2011, somewhere in the ocean between Washington and Alaska. There is evidence that food production was low off the SE Alaska coast during 2011, which may have contributed to a decrease in juvenile salmon marine survival. Primary production in the eastern Gulf of Alaska was severely depressed during 2011, which was demonstrated in anomalously low chlorophyll-a levels, an overall lower specific phytoplankton growth rate, and a greater biomass of pico-plankton cells (Strom et al., 2016). An atypically high abundance of salps in the eastern Gulf of Alaska during summer of 2011 is also believed to have contributed to low levels of phytoplankton, as these animals are capable of filtering high volumes of water on a daily basis (Li et al., 2016). The abundance of Pacific cod and walleye pollock larvae collected during spring (Siddon et al., 2016), and forage fishes captured in our survey during summer were also low (McGowan et al., 2016), further suggesting that 2011 was a year of low productivity in the Gulf of Alaska. These observations appear to have biological significance, as the 2013 spring-run Chinook salmon adult return to the Columbia River (counted at Bonneville Dam) was the lowest of the 4 relevant years of adult return data that are currently available. Taken together, it is clear that combined data from surveys conducted along the North American coastal migration pathway can provide insights into stock-specific and inter-annual variation in juvenile Chinook salmon survival.

Five years of data are insufficient to conclusively identify a relationship between juvenile relative abundance off the SE Alaska coast and subsequent returns of adult spring-run Chinook salmon to the Columbia River. There are fisheries that may also impact this relationship, however, our data reveal a pattern that supports the generally held conclusion that Chinook salmon marine survival is mostly determined within the first few months after ocean entry (Kareiva et al., 2000; Greene et al., 2005). In addition, this pattern suggests that the SE Alaska coast is a region where the marine survival of juvenile Columbia River Chinook salmon can be assessed, two years prior to their return to the river as adult fish.

Further genetic sampling of Columbia River Chinook salmon off the SE Alaska coast is clearly warranted. Our SE Alaska coast surveys in July successfully sampled Columbia River yearlings only a few weeks after they entered the ocean. Recoveries of tagged Columbia River juveniles show a subsequent dispersal of these fish westward (Healey, 1991; Regional Mark Processing Center, Pacific States Marine Fisheries Commission, www.rmpc.org/). Genetic investigations of Columbia River stocks later in their first ocean summer should therefore be conducted in offshore areas off the SE Alaska coast and in the Bering Sea. However, our single autumn survey, conducted in September 2013, shows that Columbia River Chinook salmon yearlings should also be sampled off the SE Alaska coast after summer. Approximately 81% of the fish in those samples were yearlings from Snake River and Mid/Upper Columbia River spring-run populations. This result is significant because yearlings from those sources have the fastest ocean dispersal rates and have rarely been caught in autumn surveys along the Pacific coast, presumably having all moved offshore (Fisher et al., 2014). Our data, though preliminary, reveal that at least some spring-run Chinook salmon from the interior Columbia River basin may delay their offshore migration and remain in coastal habitats into autumn.

Columbia River Chinook salmon research off the SE Alaska coast should incorporate analyses beyond the basic descriptions that we present here. Genetic data can be combined with other data to assess variability in factors such as diet (Daly et al., 2012), disease (Jacobson et al., 2008; Sandell et al., 2015), growth (Tomaro et al., 2012), and survival (Miller et al., 2014). It is critical that these studies employ genetic stock identification methods that discriminate among the diverse Columbia River Chinook salmon stocks. Marine life history, and performance and survival information that provide ESU-level insights will provide the most support for management and recovery efforts for this commercially and culturally important species (Ford et al., 2015).

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